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Ambient Noise Surface Wave Reverse Time Migration for Fault Imaging

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Key Points:

- A novel Transmitted Surface Wave Reverse Time Migration method is proposed to directly image the sharp fault discontinuities
- The reconstructed surface waves from seismic noise cross-correlations are used to image fault planes with the seismic imaging principle
- The proposed method opens up new ways to utilize the seismic ambient noise for imaging subsurface fault zone structures

Supporting Information:

- Supporting Information S1

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Abstract Ambient noise tomography (ANT) has become a routine method to image shear-wave velocity variations at different scales. However, ANT can only resolve smooth velocity variations and cannot give high-resolution images of the subsurface discontinuities. Here we propose a novel Transmitted Surface Wave Reverse Time Migration (TSW-RTM) method to directly image fault planes using the empirical surface wave Green's functions retrieved from ambient noise. Our proposed method can directly utilize the reconstructed surface wave waveforms from seismic noise cross-correlations to image the sharp fault discontinuities with the seismic imaging principle. Compared to the conventional ambient noise tomograms, the migrated images from our proposed method delineate fault planes with much higher resolution and certainty for interpretations. The synthetic tests with different fault zone models demonstrate the effectiveness and accuracy of our novel fault imaging method. The proposed TSW-RTM method is also applied to image two different segments of the Tanlu fault zone (TFZ) in eastern China, using the surface waves retrieved from the ambient noise data recorded by dense seismic linear arrays deployed across the TFZ. Our ambient noise imaging results are consistent with previous studies in the same areas regarding the fault plane positions and dips. Both synthetic tests and real data applications demonstrate that our method is very useful for fault characterizations in areas where active and passive seismic data are not available. Our proposed TSW-RTM method opens up new ways to utilize the seismic ambient noise for imaging subsurface fault zone structures.

1. Introduction

Faults are basic components of subsurface structures that represent discontinuities in physical properties. In addition to being the controlling factor of major tectonic earthquakes (e.g., Angelier, 1990; Martin et al., 2016; Sibson, 1983), subsurface faults also play an essential role in natural resources, including hydrocarbon accumulation and formation of oil and gas reservoirs (Aydin, 2000), aggregation of metallic ore deposits (e.g., Goldfarb et al., 2005; Spencer & Welty, 1986), and generation of geothermal reservoirs (Chen & Huang, 2015). Therefore, detailed characterization of subsurface faults is crucial for understanding tectonic activities and for exploring natural resources.

Many geophysical approaches for fault delineation have been proposed in the past decades, such as gravity, magnetic, electromagnetic, and seismic methods (e.g., Barbosa et al., 2007; Benson, 1995; Lyatsky et al., 2004; Wijns et al., 2005; Wu et al., 2019). Among them, seismic imaging using active seismic sources is one of the most extensively used methods to characterize fault structures in the subsurface. For instance, the Kirchhoff migration method was applied for imaging the San Andreas fault using scattered P- and S-waves from active seismic sources through the vertical seismic profiling (Chavarria et al., 2003). With the offshore active-source reflection data, J. Li et al. (2015) adopted the poststack time migration method to image the boundaries of the subducted sediments in the Alaska subduction zone for characterizing the fault structure and seismogenic behavior. Gao et al. (2016) studied the structures beneath the Yarlung Zangbo suture in western Himalaya at crustal-scale via the prestack time-migrated reflection image using the two-dimensional explosive-source data. In the oil and gas industry, the reverse time migration (RTM) imaging method (e.g., Baysal et al., 1983; Chang & McMechan, 1987; Chattopadhyay & McMechan, 2008; Levin, 1984; Yoon & Marfurt, 2006) using reflected waves has gained vast popularity and has been applied routinely to build high-resolution subsurface models (e.g., Feng & Schuster, 2017; Yang et al., 2019; Y. Zhang

et al., 2014). RTM using transmitted waves has also been tried to image steep boundaries between a salt body and surrounding sediments (e.g., Anderson & Marcinkovich, 2006; McMechan et al., 1988; Willacy & Kryvohuz, 2019).

In many prospecting areas, however, active seismic surveys may be either logistically prohibitive or too expensive to conduct. Instead, passive seismic sources such as local earthquakes can also be used for characterizing fault zone structures through seismic traveltime tomography method (e.g., Tselentis et al., 2011; H. Zhang et al., 2009a), fault head wave analysis (Ben-Zion & Malin, 1991), and fault trapped wave analysis (Y. G. Li et al., 1990). To further improve the fault imaging resolution, seismic migration based on passive seismic sources has also been used in different areas (e.g., Artman, 2006; Nakata, 2018; Reshetnikov et al, 2010; H. Zhang et al., 2009b). Recently, Lin et al. (2020) proposed a source-independent elastic reverse-time seismic imaging method to delineate hydraulic fractures using converted waves originating from the fractures or layer interfaces.

However, oftentimes the study areas are tectonically inactive and thus have no or very few earthquakes. The lack of active and passive seismic data hinders the application of those well-developed seismic imaging methods in revealing fault structures in the subsurface. Under the circumstances, ambient noise interferometry is an alternative approach for subsurface structure inference, without relying on active seismic sources or passive earthquakes (e.g., Bussat & Kugler, 2011; Gu et al., 2019; Kirschner et al., 2018; Mordret et al., 2014; Shapiro et al., 2005). Using the cross-correlation and stacking techniques, the seismic empirical Green's functions (EGFs), which are mainly comprised of strong surface waves and relatively weak body waves are retrieved from seismic ambient noise, as an approximation of the Green's functions between a pair of stations. For ambient noise imaging using the retrieved body waves, Draganov et al. (2009) used the poststack time migration method to obtain the body-wave reflection images of the structures in the near surface, and Poli et al. (2012) imaged the mantle discontinuities using the extracted body waves via stacking techniques in the slowness-time domain. Nakata et al. (2015) retrieved P diving waves from seismic ambient noise recorded by a 2D dense network with 2,500 receivers, and then inverted for the 3D velocity structure under the network using traveltime tomography. As for the surface wave ambient noise imaging, the retrieved surface waves are converted to dispersion data, which are subsequently used in tomography to characterize the shallow or deep earth structures (e.g., Chmiel et al., 2019; Fichtner, 2014; Fichtner et al., 2016, 2017; Mordret et al., 2014; Shapiro et al., 2005; Wang et al., 2019; Yao et al., 2006). Usually, the retrieved surface waves are more energetic and reliable than the retrieved body waves due to the distribution of the noise sources (e.g., Boué et al., 2013; Draganov et al., 2009; Fichtner & Tsai, 2019; Schuster, 2016; Shapiro & Campillo, 2004; Stehly et al., 2006), and the signal-to-noise ratio (SNR) for the resulting seismic images obtained with surface waves is higher.

Like seismic traveltime tomography, surface wave ambient noise tomography (ANT) yields a smoothed S-wave velocity model which lacks clear structural delineation, posing challenges for detailed structural interpretation. To take advantages of the energetic surface waves, we propose a novel ambient noise Transmitted Surface Wave Reverse Time Migration (TSW-RTM) method to image subsurface faults in a higher resolution, across which velocity varies considerably due to lithology changes. In the proposed TSW-RTM method, we aim to better delineate fault planes which separate low-velocity sediments from high-velocity metamorphic or igneous rocks, as shown in the case of the Tanlu fault in eastern China (Gu et al., 2019). A simplified fault zone model composed of high-velocity metamorphic or igneous rocks (green area), low-velocity surrounding sediments (bright-yellow area), and a crystalline basement (pink area) is shown in Figure 1. Both the San Andreas fault in the Western U.S. and the Tanlu fault zone (TFZ) in eastern China have high velocity contrast across the fault (e.g., Fattaruso et al., 2016; Thurber et al., 2004; Yeats, 2012).

In our proposed method, the surface wave waveforms extracted from ambient seismic noise (e.g., Shapiro & Campillo, 2004; Yao et al., 2006) are directly used to image fault planes. Compared with the conventional ambient noise tomographic methods, our method yields higher-resolution images of the target fault planes. Since our method takes advantage of the strong surface wave energy instead of body waves retrieved from seismic ambient noise, it is very robust. The proposed TSW-RTM method can yield a valid image of a fault plane down to the depth which the utilized surface waves are sensitive to. Another merit of the proposed TSW-RTM method is that only rough a priori information on the fault discontinuities is required in building the migration velocities via the velocity flood technique and designing the acquisition geometry for migration.

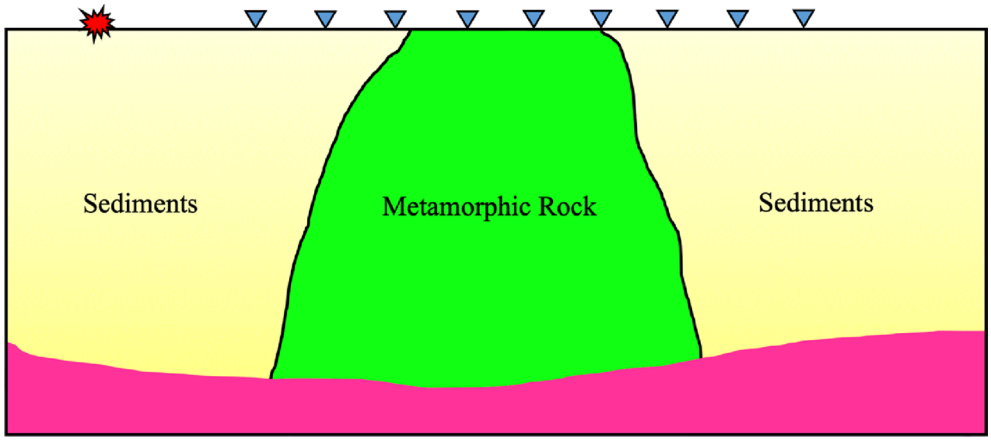


Figure 1. A simple fault model with seismic survey layout. The high-velocity metamorphic rocks are surrounded by low-velocity sediments, with underlying crystalline basement. The black lines represent fault planes we aim to image.

This study is organized as follows. First, we show the theory and workflow of the proposed TSW-RTM, and then discuss the ambient noise interferometry for surface wave retrieval; second, we perform synthetic tests with various fault models to validate the proposed TSW-RTM, and investigate the impact of migration velocity uncertainties on the fault plane images; third, our method is applied to image several major faults of the TFZ near the cities of Hefei and Suqian in eastern China, using the surface waves retrieved from seismic ambient noise recorded by dense linear arrays across the target faults; finally, we analyze the factors which control the lateral imaging resolution and the maximum imaging depth.

2. Method

2.1. TSW-RTM Based on the Velocity Flood Technique

Before we introduce the proposed TSW-RTM method, we briefly review the conventional RTM method which has been broadly used in the oil and gas industry in the last decade. Based on the two-way wave equation, the source wavefield propagates forward in time, and the receiver wavefield is extrapolated backward in time, and a zero-lag cross-correlation imaging condition (Claerbout, 1971) is applied to the forward and backward wavefields:

$$\boldsymbol{I}(\mathbf{x}) = \int_0^T s(\mathbf{x},t)\, r(\mathbf{x},t)\, dt, \tag{1}$$

where $s(\mathbf{x}, t)$ and $r(\mathbf{x}, t)$ denote the forward wavefield from a source and the backward wavefield from the receiver at position $\mathbf{x}$ and time t, respectively; T represents the total propagation time; I denotes the migrated image. The imaging condition states that the migrated image is generated from constructive interferences wherever the source and receiver wavefields coincide spatiotemporally. Stacking many pairs of sources and receivers can further enhance the SNR and the resolution of the image. RTM can image complex structures of varying dips in the subsurface since the source and receiver wavefield propagate in all possible directions (e.g., Chang et al., 2011; Tan & Huang, 2014).

Rather than trying to resolve structures with all different attributes like the conventional reflection RTM, the proposed TSW-RTM, which is based on the velocity flood technique (Willacy & Kryvohuz, 2019), is designed specifically to image fault planes with high-velocity contrast. We consider a simple model as shown in Figure 2a, which is composed of some low-velocity sediments and high-velocity metamorphic rocks separated by a fault plane (denoted by the black line in the middle). Velocity flood refers to extending the velocity model from one side of the fault boundary all the way to enclose the receivers or sources on the opposite side beyond the boundaries, and a correct velocity model is intentionally replaced by a pair of incorrect velocity models (Figures 2b and 2c). By replacing the original velocity model which contains a sharp velocity contrast with the flood velocity models which bear incorrect velocity structures

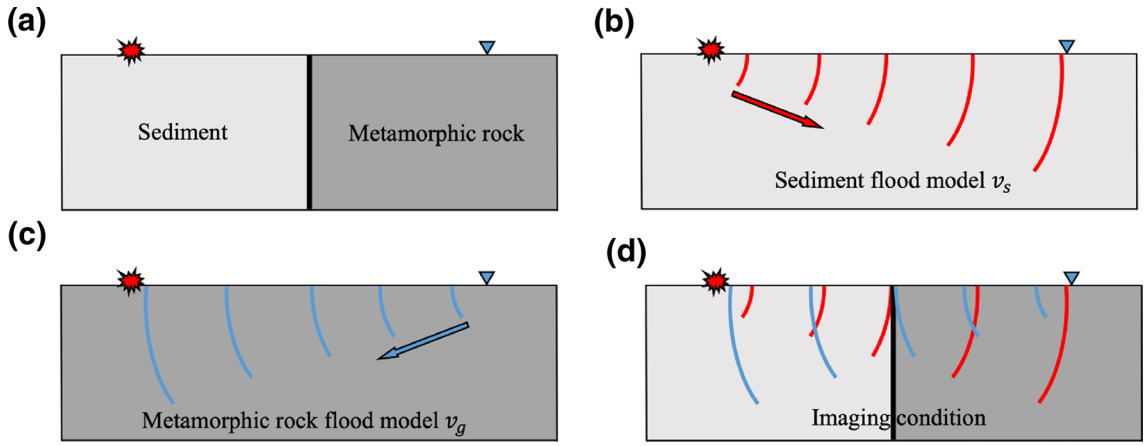


Figure 2. (a) The velocity model composed of the sediment (left) and the metamorphic rock (right); (b) the forward wave-field propagation from the source using the sediment flood velocity; (c) the backward wave-field propagation from the receiver using the metamorphic rock flood velocity; (d) the cross-correlation of the forward and backward wavefields.

starting from where the replacement takes place, the forward- and backward-propagated wavefields have spatiotemporal constructive interference only at the true fault discontinuity with unknown position. The constructive interference is gradually lost away from the fault discontinuity in either direction. It should be emphasized that accurate a priori information regarding the actual positioning of the fault plane is not required in the replacements, that is, the velocity flood can take place certain distance away from the true discontinuity. Details of the velocity flood will be further elaborated in the synthetic tests and in the real data examples.

The proposed TSW-RTM method takes the following three steps: (1) simulate the wavefield in the true velocity model to obtain the recorded shot gathers, or in the real data cases, construct the virtual shot gathers from the ambient noise data with cross-correlations; (2) excite the forward-propagating wavefield from the source situated on top of the sediments in the sediment flood velocity model v_s (Figure 2b), where the high velocity metamorphic rock on the other side of the fault is entirely replaced by the extended sediments (sediment flood); (3) excite the time-reversed wavefield from the receiver (i.e., time-reversing the recorded data at the receivers, and then injecting the processed data at the receiver positions) situated on top of the metamorphic rock in the metamorphic flood velocity model v_g (Figure 2c), where the sediment deposited at the opposite side of the fault is entirely replaced by extended metamorphic rocks (metamorphic rock flood); (4) correlate the forward- and backward-propagated wavefields from steps (2) and (3) by applying the zero lag cross-correlation imaging condition to obtain the migrated image of the fault plane, as illustrated in Figure 2d. In such a way, the forward and backward wavefields only spatiotemporally coincide at the actual velocity discontinuity with the flooded velocity models. In our study, a standard staggered-grid finite difference elastic solver is used to propagate the wavefields (Levander, 1988), with the W-AFDA scheme adopted for modeling the surface waves (Kristek et al., 2002).

The process above can be explained by the following equation (e.g., Schneider, 1978; Schuster, 2017):

$$\boldsymbol{I}\left(\boldsymbol{x} \mid \boldsymbol{x}_s, \boldsymbol{x}_g\right) = \frac{A}{\left|\mathbf{x}-\mathbf{x}_g\right|\left|\mathbf{x}-\mathbf{x}_s\right|}\delta\left(t - \frac{\left|\mathbf{x}-\mathbf{x}_s\right|}{v_s} - \frac{\left|\mathbf{x}-\mathbf{x}_g\right|}{v_g}\right), \tag{2}$$

where $\mathbf{x}_s$ and $\mathbf{x}_g$ denote the source and receiver locations, respectively; v_s and v_g denote the sediment and metamorphic rock flood velocities for propagating the source and receiver wavefields, respectively; t denotes the travel time of the surface wave from the source to the receiver; A denotes the amplitude of the image, which is determined by the source wavelet and the travel distance of the surface wave. The image $\boldsymbol{I}(\boldsymbol{x}|\boldsymbol{x}_s,\boldsymbol{x}_g)$ forms an isochrone surface (Savard, 2001) for a specific source and receiver combination, and the stacked result from all wavefields derived from zerolag correlations is the desired image characterizing the velocity discontinuities. However, unlike body waves, surface waves (Rayleigh wave in our case) only

propagate along the free surface. With the plane-wave assumption at the far wavefield, the analytical expression for the surface wave at a single frequency ω is (Shapiro, 2019):

$$u_s\left(\mathbf{x}_{fs},z,t,\omega\right) \approx \xi\left(z,\omega\right)\exp\left[i\omega\left(t-\frac{\mathbf{x}_{fs}}{C\left(\omega\right)}\right)\right], \tag{3}$$

where $\xi\left(z,\omega\right)$ denotes the sensitive kernel describing the amplitude decay along depth, $C\left(\omega\right)=\omega/k\left(\omega\right)$ denotes the phase velocity, and $\mathbf{x}_{fs}$ denotes the position of the wave front on the free surface. By applying the cross-correlation imaging condition to the surface waves of a single-frequency, the migrated image is:

$$\boldsymbol{I}\left(\boldsymbol{x}_{fs},z,\omega\right)=\xi_s\left(z,\omega\right)\xi_g\left(z,\omega\right)\delta\left(t-\frac{\left|\mathbf{x}_{fs}-\mathbf{x}_s\right|}{C_s\left(\omega\right)}-\frac{\left|\mathbf{x}_{fs}-\mathbf{x}_g\right|}{C_g\left(\omega\right)}\right), \tag{4}$$

where $\xi_s\left(z,\omega\right)$ and $\xi_g\left(z,\omega\right)$ are the sensitivity kernels of the surface waves excited from the source and the receiver, respectively. The final image consisting of all frequencies therefore is:

$$\boldsymbol{I}\left(\mathbf{x}_{fs},z\right)=\int_0^\infty \boldsymbol{I}\left(\mathbf{x}_{fs},z,\omega\right)d\omega. \tag{5}$$

Due to the dispersion effect, surface waves of different frequencies propagate with different velocities and are sensitive to different depths in a vertically changing structure (Yanovskaya et al., 2012), which means that the dispersion of the surface wave is key to constructing the image at different depths. Stacking the contributions from different source-receiver pairs can enhance the image of the fault plane and suppresses the artifacts, which is demonstrated in the next section. In summary, the proposed imaging procedure is similar to the common reflection RTM, except that: (1) transmitted surface waves propagating along the free surface are used instead of the reflected body waves; (2) the flood velocities are used as the migration velocities in the propagation of the forward and backward wavefields, to correctly form the migrated image.

It should be noted that the choice of the source and receiver is arbitrary and can be swapped in the proposed imaging method. If the source location is deployed on the metamorphic rock side and the receiver on the sediments side, the only difference is that the forward wavefield is propagated in the metamorphic rock flood velocity and backward wavefield in the sediment flood velocity, while the other imaging steps remain unchanged. The proposed method does not require specific source-receiver orientation as long as they are situated at the two different sides of the target fault plane (please refer to Figures S1 and S4 in supporting information for comparison). Therefore, in the following numerical tests, we only consider the situation where the sources are deployed on the sediment side and receivers on the metamorphic rock side.

2.2. Using Surface Waves Retrieved From Ambient Noise Interferometry

The theory of seismic noise interferometry has been covered by many previous studies (e.g., Delaney et al., 2017; Nakata et al., 2015; Shapiro, 2019; Shapiro et al., 2005). The noise cross-correlation theorem states that the time derivative of the cross-correlation of seismic ambient noise recorded by a pair of receivers converges to the Green's functions between the two positions, which is:

$$\frac{dC_{AB}\left(t\right)}{dt}=-\hat{G}_{AB}\left(t\right)+\hat{G}_{BA}\left(-t\right)\approx -G_{AB}\left(t\right)+G_{BA}\left(-t\right). \tag{6}$$

where $C_{AB}(t)$ is the cross-correlation functions (CCFs) of the ambient noise wavefields recorded at locations A and B; $\hat{G}_{AB}\left(t\right)$ and $G_{AB}(t)$ are the empirical and the actual Green's functions from location A to B, respectively; $\hat{G}_{BA}\left(-t\right)$ and $G_{BA}(-t)$ are the time-reversed empirical and actual Green's functions from B to A, respectively. Previous studies (e.g., Draganov et al., 2009; Wapenaar, 2004) show that a virtual common shot gather can be retrieved from noise cross-correlations if sufficiently long records are used. It should also be mentioned that in the ambient noise processing receivers can also act as virtual sources, thus the source and receiver positions in TSW-RTM can be swapped with the ambient noise data. Thus, any receiver in an

array can be treated as a virtual source emanating a wavefield characterized by the corresponding noise correlation functions to the rest receivers in the same array.

3. Numerical Tests

3.1. Tests of Different Fault Models

To investigate the effectiveness and accuracy of the proposed TSW-RTM, we first consider a synthetic fault zone model illustrated in Figure 3a, which resembles the typical fault zone structure in the southern segment of the TFZ (Gu et al., 2019). In this model, the fault zone comprising of high velocity metamorphic rock with S-wave velocity of 3,000 m/s is situated in the middle, surrounded by sediments with S-wave velocity varying vertically from 980 to 1,800 m/s. In the following synthetic tests, the V_p/V_s ratio is assumed to be 1.73, and the density is calculated according to the Gardner's relationship (Gardner et al., 1974). It should be noted that the velocity of the metamorphic rock is assumed to be constant only for the sake of discussions. Our method is indeed effective for fault zones with varying velocities (refer to Figure S6 in supporting information). The fault zone spans 8 km in the near surface and further widens with depth. We aim to image the two faults which separate the high-velocity fault zone from the low-velocity sediments. Two groups of sources (red and blue stars) are deployed on the surface for imaging the left and right faults, respectively, and 16 receivers are situated immediately above the fault zone (black triangles). The red and blue dashed lines indicate the positions where we perform the velocity flood to build the migration velocities for imaging the left and right fault flanks, respectively. To image the left flank, the forward modeling velocity is built by extending the sediment velocities from the left red dashed line to the right edge of the model (Figure 3b), while the backward modeling velocity is obtained by flooding the metamorphic rock from the second red dashed line to enclose the sources on the left indicated by the red stars (Figure 3c). To image the right flank, a similar process is undertaken (Figures 3d and 3e). In this test, 16 sources on the left (red stars) are excited separately with a 1 Hz Ricker wavelet for imaging the fault boundary on the left. The same procedure goes for the 16 sources on the right (blue stars) for imaging the fault boundary on the right. For fault imaging in both cases, the forward wavefields propagate only in the sediment flood velocity model, whereas the time-reversed wavefields only in the metamorphic rock flood velocity model, respectively. The stack of the cross-correlations from all source and receiver pairs in the two independent acquisitions lead to the final grayscale image as shown in Figure 3f, which overlays on the reference velocity model in the background. Figure 3f shows that the largest amplitudes of the image are well co-located with both the left and right flanks of the fault zone. That is, from the imaging perspective the fault planes are correctly imaged in terms of their positions and variations of dipping angles from shallow to deep. Surface waves are dispersive: those with shorter wavelengths and higher frequencies mainly propagate at shallower depths, whereas those with longer wavelengths and lower frequencies mainly propagate at greater depths. Therefore, the horizontal resolution of TSW-RTM is inversely proportional to the depth, which explains why the shallower segments of the boundaries are better resolved while the resolution gradually decreases with the depth. We also perform amplitude normalization along the horizontal direction to visually enhance the image in the deeper part where the surface wave amplitude is relatively weak (refer to Figure S7 in supporting information for the enhancement effect).

To further validate our method, another four intricate synthetic fault zone models are tested where the fault planes are featured by varying dip angles, and shapes, as shown in the left panels of Figure 4. From models (a–g) in Figure 4, the dip angles of the faults become less and less steep. The layouts of the acquisitions are the same as those in the previous test. The imaging results are shown in the right panels of Figure 4. The maximum amplitudes of the migrated image (Figure 4b) for the first narrow fault zone model is positioned precisely at the steep-dipping fault discontinuities. The fault zone centered slightly to the left shown in Figure 4c is less steep but more complicated on the left side. The corresponding imaging result as shown in Figure 4d clearly resolves the upper part of the fault on the left, but its resolution decreases slightly with depth. A similar imaging result (Figure 4f) can be found for the third model centered slightly to the right as shown in Figure 4e, which has a complicated fault on the right. In the fourth wide fault zone model, the upper parts of both shallow-dipping faults are also clearly imaged, while the lower parts are imaged with decreasing resolution. Overall, various faults with different features are imaged with the proposed TSW-RTM correctly, while the imaging resolution of the fault planes is dependent on both the depth and the dip angles.

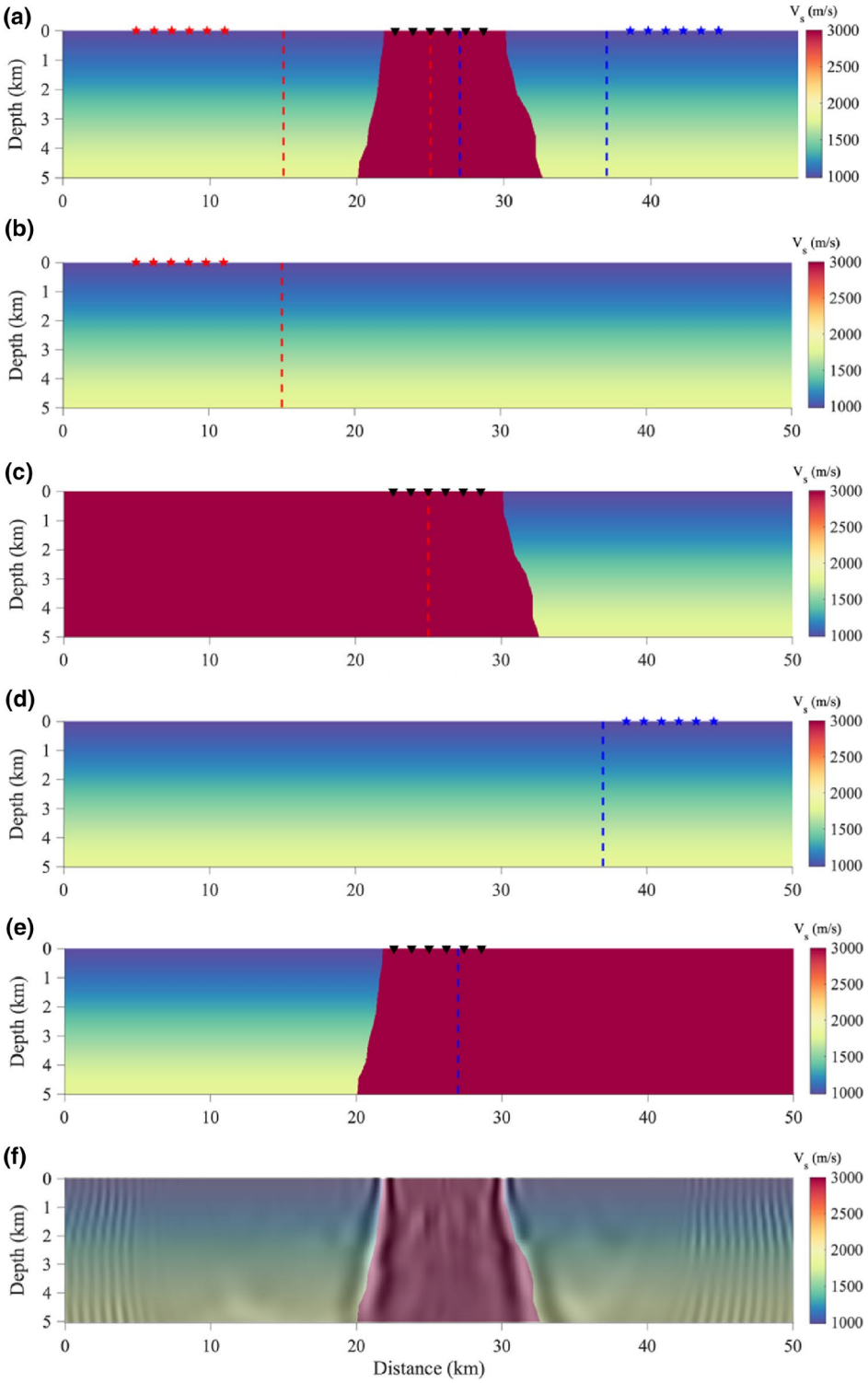


Figure 3. (a) A synthetic fault zone model. The red and blue stars represent source positions on both sides of the fault zone, and the black triangles represent the receiver positions, respectively. In the numerical tests, 16 sources and 16 receivers are actually adopted to image each flank of the fault zone, but fewer sources and receivers are plotted here for clarity. The red and blue dashed lines indicate positions where velocity flooding is performed to build the forward and backward migration velocities. (b and d) Sediments flood velocity models with sources indicated by the red and blue stars for two independent acquisitions, respectively. (c and e) Metamorphic rock flood velocity models with receivers indicated by inverted triangles for two independent acquisitions, respectively. (f) The migrated image using Transmitted Surface Wave Reverse Time Migration, which is shown in grayscale with 65% transparency, overlaying on the true velocity model shown in color.

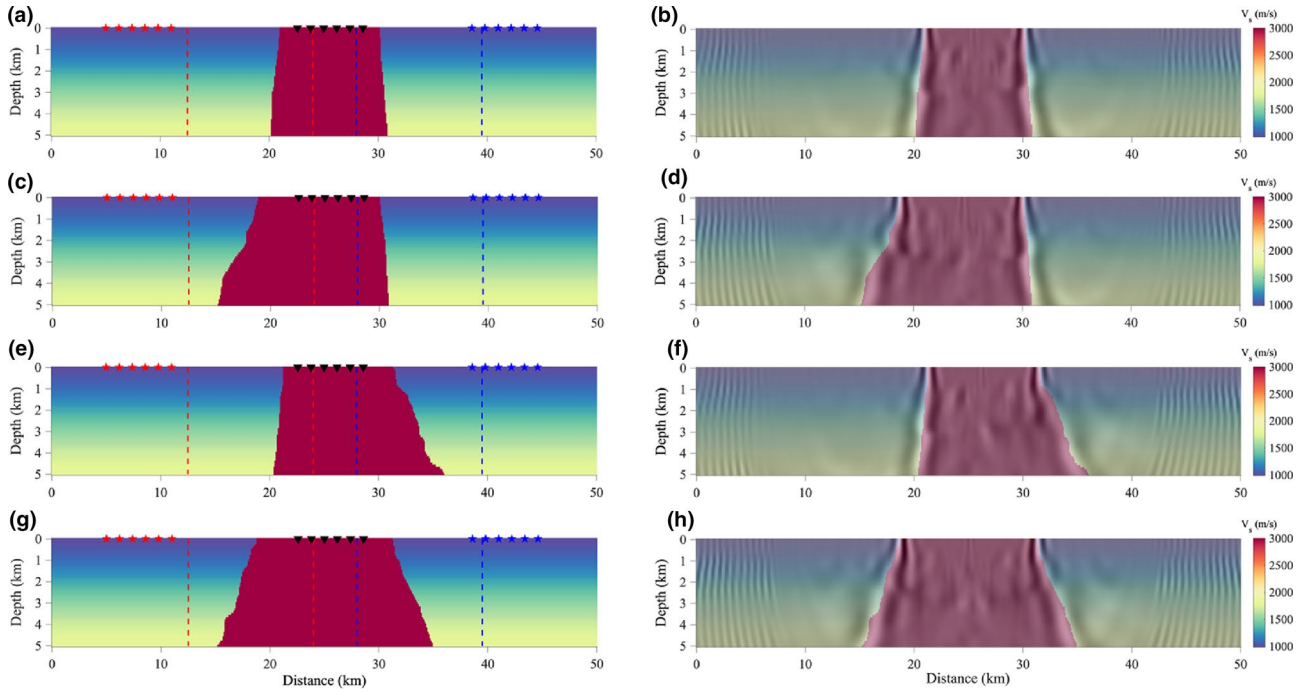


Figure 4. Four synthetic fault zone models with different dipping angles and shapes and their respective imaging result. There are 16 sources (red or blue stars) and 16 receivers used to image each flank of the fault zone. The red and blue dashed lines indicate positions where velocity flooding is performed to build the forward and backward migration velocities. (a and b) The fault zone model with two steep fault planes and the imaging results; (c and d) the fault zone model with one steep-dip fault and one moderate-dip but intricate fault on the left, and the imaging result; (e and f) the fault zone model with one steep-dip fault and one moderate-dip but intricate fault plane on the right, and the imaging result; (g and h) the fault zone model with two moderate-dip faults and the imaging result. The imaging results are shown in grayscale with 65% transparency, overlaying on the true velocity model shown in color.

The seismic acquisition should be planned such that the surface wave can provide sufficient penetration and illumination to image the target fault plane at greater depth. Due to the variations in the surface wavefield penetration, the imaging results of the fault plane may differ slightly, especially at greater depth when using different array geometries. For more imaging examples regarding the source-receiver configuration, refer to Figures S1 to S5 and related descriptions in supporting information.

3.2. Impact of the Migration Velocity

In order to obtain a well-focused migrated image, it is crucial to analyze the impact of the flood velocity on the imaging quality. We carry out four more synthetic tests to investigate the images obtained with flood velocity models with some uncertainties (Figure 5). First, we consider the case where the velocity model for metamorphic rock flood is 10% faster, which is used for back-propagating the wavefields from the receivers. It is noticed that the imaged fault planes on both sides are pushed inwards into the fault zone slightly (Figure 5a). In contrast, when the velocity model for metamorphic rock flood is 10% slower, the imaged faults (Figure 5b) are pulled outwards from the fault zone slightly. We then consider the impact of an erroneous velocity model for sediment flood. As shown in Figure 5c, when the sediment flood velocity is 10% faster, the imaged faults are pushed inwards into the fault zone noticeably. In comparison, when using a 10% slower sediment flood velocity (Figure 5d), the imaged faults are pulled outwards from the fault zone significantly. In our tests, the surface waves propagate for a longer distance in the sediments than in the metamorphic rocks in these models, thus errors of the same percentage in the sediment flood velocity model has a more pronounced influence on the final imaging quality. It is concluded that a faster metamorphic rock or a sediment flood velocity model always pushes the fault images toward the center of the fault zone, while a slower metamorphic rock or sediment flood velocity pulls the imaged faults away from the fault zone.

A similar boundary-shift pattern when using transmission RTM is also observed by Willacy and Kryvohuz (2019) when an inaccurate velocity model is used. It should be mentioned that images obtained by the

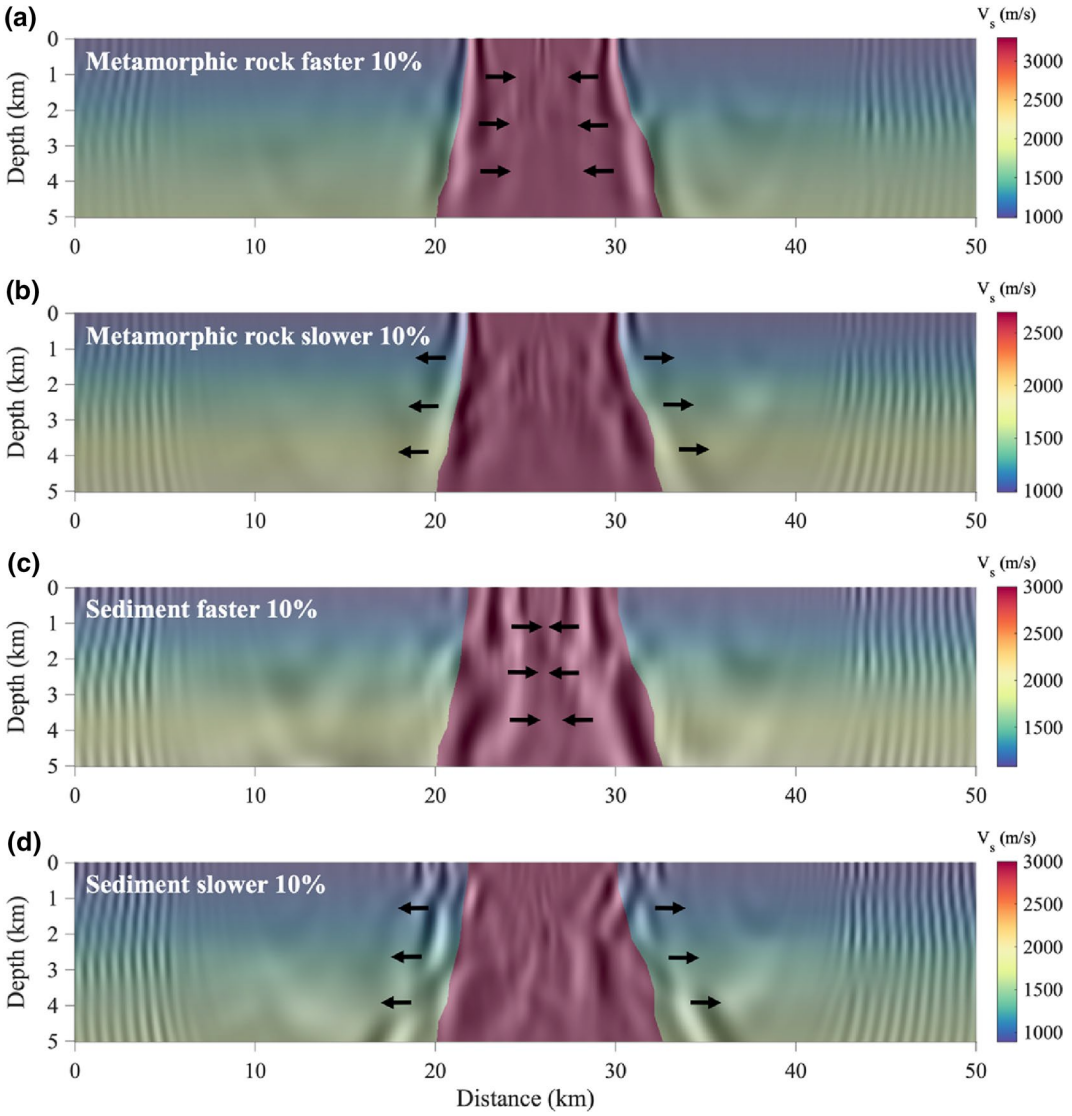


Figure 5. (a) Imaging result using a 10% higher metamorphic rock flood velocity model; (b) imaging result using a 10% lower metamorphic rock flood velocity model; (c) imaging result using a 10% higher sediment flood velocity model; (d) imaging result using a 10% lower sediment flood velocity model. The imaging results are shown in grayscale with 65% transparency, overlaying on the true velocity model shown in color.

transmission RTM and the reflection RTM shift toward the opposite directions when using an inaccurate velocity model (Willacy & Kryvohuz, 2019). That is, a higher flood velocity causes the image shift toward the center of the fault zone in the transmission RTM imaging, whereas the same erroneous velocity model shifts the fault toward the surrounding sediments in the reflection RTM imaging. Therefore, if both transmission and reflection RTM are performed, the fact that the fault image shifts toward opposite directions with an inaccurate velocity model can be utilized to cancel out mispositioning and better image the fault.

In the particular numerical experiments above, sources are situated on top of the sediments while the receivers are deployed on top of the metamorphic rocks across the fault. However, influence of the velocity inaccuracy on fault mispositioning remains the same even if the sources and receivers are swapped, according to source-receiver reciprocity. That is, in the following real data examples where the virtual shot gathers are retrieved from seismic ambient noise, in principle any receiver can be treated as the source. In practice, however, the correlation functions are sometimes asymmetrical since the noise sources are distributed unevenly. In such circumstances, one receiver is favorably chosen as the source than the other, depending on

whether the causal or the anticausal branch of the correlation function has higher SNR. The causal and the anticausal parts of the CCFs are both used in constructing the virtual shot gathers, which are located at both sides of a fault plane. We then decide how to select the overall source and receiver geometry according to the SNR of the retrieved shot gathers, with the constraint that all the selected virtual shots are located on one side of the fault plane, and receivers on the other side.

The numerical tests demonstrate that the proposed method can correctly image the fault planes with a roughly correct velocity model containing the seismic array and the fault regions. More importantly, only the flood velocity models are used in TSW-RTM, that is, only rough information of the fault position is required in those modified velocity models for imaging. This nice property largely improves the applicability of the proposed method in the field data applications, since oftentimes only the approximate location of a fault can be determined from outcrops or other available geophysical measurements. This is especially true for blind or hidden faults.

4. Applications in Imaging the Southern Segment of the TFZ

4.1. Background of the Study Areas

Our study is focused on imaging several major faults near the cities of Hefei and Suqian along the TFZ. As the largest fault zone in China trending in the north by northeast direction, the Tancheng-Lujiang (Tan-lu) fault extends over 2,400 km from the Lujiang City in the Anhui Province to the Tancheng City in the Shangdong Province, and beyond. In spite of being the largest fault zone located in the eastern part of the Eurasia continent, the TFZ has been seismically inactive in the past 300 years, with the most recent major earthquake occurring in the 17th century (Bem et al., 2020). Though many studies have been carried out in the past few decades, several key questions regarding the fault zone remain unanswered or controversial geologically (e.g., Liang, 2018; Zhang et al., 2008; Zhu et al., 2011). Gilder et al. (1999) investigate the tectonic evolution pattern of the TFZ from middle Triassic to early Cenozoic using paleomagnetic samples. With many field surveys and the optically stimulated luminescence dating method, Liu et al. (2015) study the Quaternary faulting of the TFZ near the Suqian City (area A in Figure 6a). Active seismic surveys were also conducted in both areas to characterize the fault structures with seismic refractions and reflections (Qin et al., 2018, 2020). Gu et al. (2019) investigate the shallow crustal structures adjacent to area B to the north (Figure 6a) using a 3D ANT. Previous studies reveal that the TFZ in both the Hefei and Suqian segments contain high-velocity blocks of Precambrian ultrahigh pressure metamorphic rocks in the shallow crust, and the TFZ may act as a channel for magma intrusion. Across the F15 fault to the west in area B (Figure 6c), the lithology changes from high-velocity metamorphic rocks to low-velocity quaternary sediments. More details on the geology of the TFZ is beyond the scope of this paper.

In the conventional ambient noise surface wave tomography (Gu et al., 2019) and active source seismic refraction tomography (e.g., Qin et al., 2018, 2020), it is difficult to determine the accurate position of a fault or to delineate the subtle variations in dip angles along its surface, since a smoothing constraint is inevitably applied and the tomogram is forfeited sharpness thereof. In order to obtain high-resolution subsurface fault structures, we apply the proposed method to image the major faults in TFZ using surface waves retrieved from ambient seismic noise.

4.2. Data Acquisition, Preprocessing, and Inversion for the Migration Velocity Model

The seismic ambient noise data were recorded by a 2D linear array in area A and another linear array in area B, respectively (Figure 6a). In area A near Suqian, a 60 km linear array (red triangles) covering four late Pleistocene faults (F1, F2, F3, and F4) and one Quaternary fault (F6) was deployed to record seismic ambient noise. This linear array was deployed for a period of 30 days with 40 EPS-2-M6Q 5 s nodes (red triangles), as shown in the zoomed-in view in Figure 6b. In the same area, a 96 km long, active-source seismic profiling was conducted along the purple line as shown in Figure 6a, part of which is collocated with the passive ambient noise array. In area B, a shorter but denser linear array spanning 22 km was deployed across the F15 fault (Figure 6c), located near the Chao Lake southeast of Hefei city to record seismic ambient noise. This linear array was deployed for a period of 35 days, with 45 Zland-3C 5 Hz nodes with an average interstation distance of 500 m.

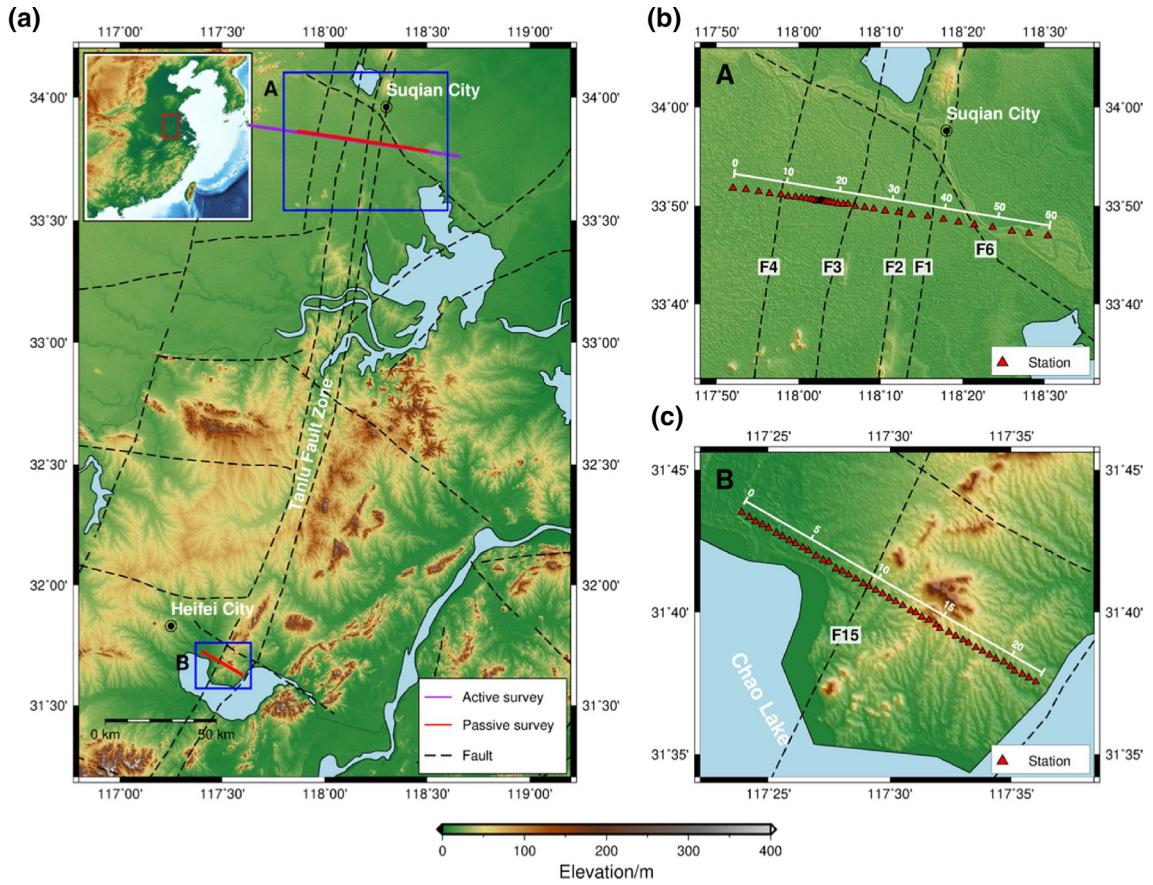


Figure 6. (a) Two linear arrays (red lines) deployed in the southern segment of Tanlu fault zone. The northern array in area A is near the city of Suqian and the southern array in area B is near the city of Hefei, respectively. The surface fault traces are denoted in dashed lines. In area A, the active-source seismic profile used for comparison is denoted by the purple line. The zoomed-in view of the survey areas A and B in the blue boxes are shown in (b and c), respectively, where the red triangles represent individual nodes.

Following the standard ambient noise processing procedure (Bensen et al., 2007), all the vertical component data were first cut into hourly records and decimated at a sampling rate of 50 Hz for data sets from areas A and B (data sets A and B for short). The mean and trend of the data were then removed, followed by band-pass filtering applied to data set A between 0.5 and 10 s and data set B between 0.1 and 5 s. To suppress the influence of impulsive energy from earthquakes, we also performed spectral whitening and multiple-band temporal normalization for both data sets. After those steps, we calculated the ambient noise cross-correlation between two stations for each hourly data with lag-time ranging from −100 to 100 s for data set A, and −60 to 60 s for data set B. All the hourly CCFs from each station pair were stacked by using the normalized linear stacking method over 30 and 35 days for data sets A and B, respectively. We averaged the positive and negative lags and finally obtained EGFs from CCFs using the Hilbert transform (Yao et al., 2011). We then used the image analysis method by Yao et al. (2006) and Yao et al. (2011) to extract the fundamental mode Rayleigh wave group and phase velocity dispersion curves. In total, from data set A we obtained 66 group velocity dispersion curves in the 0.5–8 s period band, and 34 phase velocity dispersion curves in the 0.5–6.5 s period band; for data set B, we obtained 331 group velocity dispersion curves in the 0.2–5 s period band, and 262 phase velocity dispersion curves in the 0.2–5 s period band.

The stacked CCFs, after removal of the instrument response for all the receiver pairs in areas A and B, are shown in Figures 7a and 7d in the band of 1.0–5.0 s, respectively. For data set B, the asymmetry in the noise correlation functions is caused by the uneven distribution of the noise sources. The low-frequency signals (>1.0 s) probably originated from the swell generated seismic noise, for example, by the Chao Lake, the East China Sea, or the Yellow sea. While the anthropogenic activities in the city of Hefei located to the north-west of the linear array may only contribute to the high-frequency components (<1.0 s). For the imaging

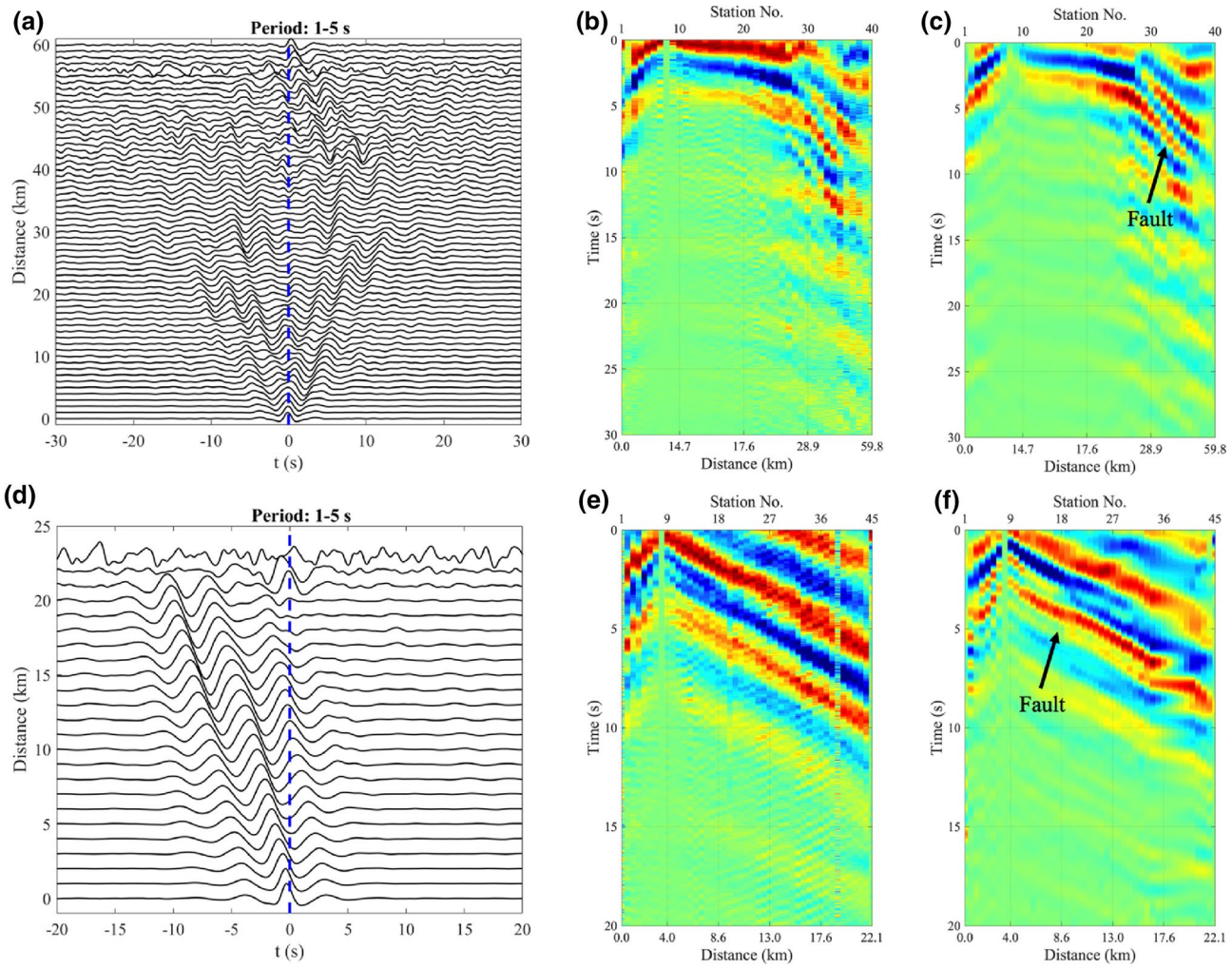


Figure 7. The cross-correlation functions (CCFs) and virtual shot gathers from seismic ambient noise interferometry for two linear arrays in areas A and B. (a) The CCFs obtained for the linear array in area A. (b and c) give a raw virtual shot gather and the processed (time-derivative, F-K, and band-pass filtering) shot gather for area A. The corresponding plots for area B are shown in (d–f), respectively.

purpose, the preliminary virtual shot gathers (Figures 7b and 7e) are then extracted from the CCFs. After preprocessing, including taking the time-derivative (Equation 6), performing the F-K filtering and band-pass filtering (1.0–5.0 s), we obtain the final virtual shot gathers (Figures 7c and 7f), which are back-propagated from the receivers in TSW-RTM. According to the sensitivity kernel analysis for the group and phase velocities for the Rayleigh wave, the retrieved surface waves are sensitive at least down to 6 km in depth for data set A in the Suqian area, and to 4 km in depth for data set B in the Chaohu area. Refer to Figure S8 in supporting information for more details. In Figure 7c, it is noticed that the continuity of the waveforms is interrupted at around 40 km. It seems there is a half-cycle advance of the surface waves, followed immediately by a change in the apparent velocity of the surface wave. It may suggest the surface wave in this virtual shot propagates into a high velocity zone at around 40 km, then after propagating for a short distance it reenters another low-velocity zone. In comparison, discontinuity and change of the apparent velocity in the virtual shot gathers seem to be less severe in the preprocessed shot gather from area B (Figure 7e), though it is still observable that there is a minor change in the apparent velocity around station 18.

The surface-wave group and phase velocity dispersion data extracted from the CCFs are inverted simultaneously for the smooth S-wave velocity models used for migration with the direct surface wave inversion method (Fang et al., 2015). The P-wave velocity is set to be 1.73 times the S-wave velocity, and density models are inferred using the empirical relations (Gardner et al., 1974). Since the velocity of the surface wave is

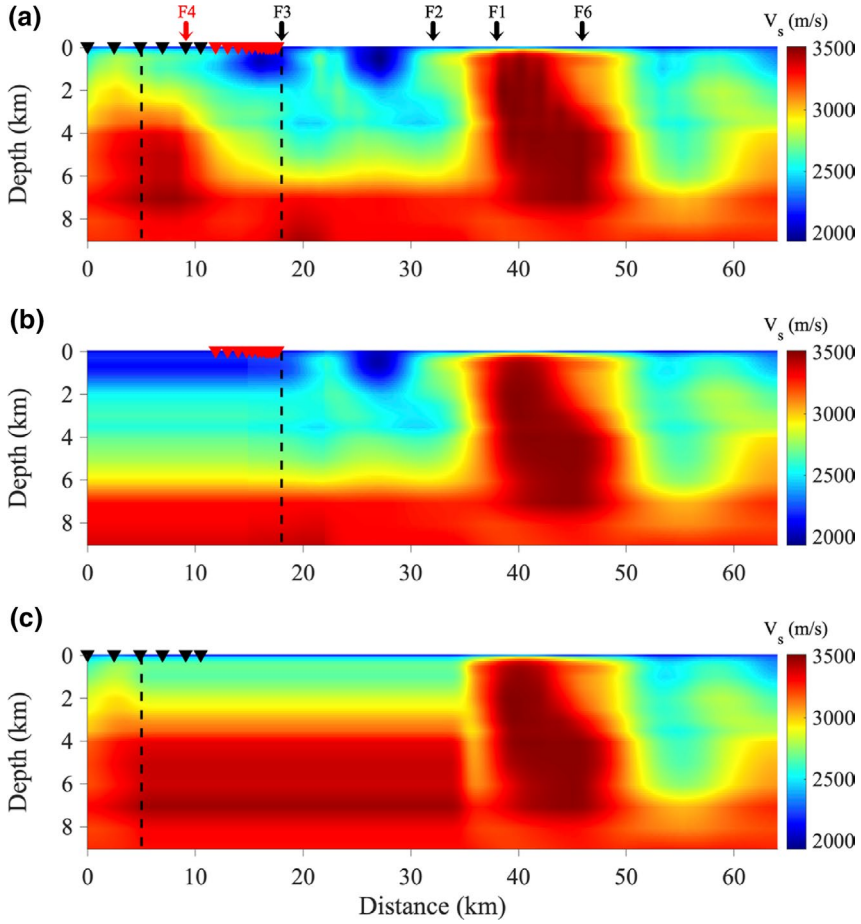


Figure 8. (a) The S-wave velocity model from the ambient noise tomography along the linear array in area A where the surface positions of the F1–F6 faults are marked by the red or black arrows. The surface location of the F4 fault is situated between the two dashed lines. The black and red inverted triangles represent the 6 receivers and 14 virtual sources used in imaging the F4 fault, respectively; (b) the sediment flood velocity used to forward-propagate the source wavefield; (c) the metamorphic rock flood velocity used to back propagate the time-reversed wavefield.

mostly sensitive to the S-wave velocity (Fang et al., 2015), the approximations only cause very minor influence on the quality and accuracy of the fault imaging results. The tomography result and the flood velocity models for imaging three different faults in area A are shown in Figures 8–10, respectively. Figure 11 shows the tomography result and the flood velocity models for imaging the F15 fault in area B. The characteristics previously observed in the virtual shot gathers are consistent with the velocity tomographic results as shown in Figures 8a and 11a.

To further validate our tomographic results, we generate synthetic shot gathers for comparison with the observed ones using the tomographic models for areas A and B (Figures 8a and 11a) in Appendix A. In general, it is found that the observed and synthetic shot gathers share similar patterns and variations in the apparent velocity.

4.3. Fault Imaging Results With TSW-RTM

In the inverted S-wave velocity model in area A (Figure 8a), two high-velocity zones can be found between distance 0 and 10 km and between distance 35 and 50 km, respectively. Two low-velocity sedimentary basins can also be found between distance 10 and 35 km, and between distance 50 and 60 km. Three potential faults, namely the F4, F1, and F6 faults can be deduced based on the velocity contrasts at around distance 10, 35, and 50 km, respectively, and the contrasts correlate well with surface fault positions. The velocity contrasts at the other three surface-identified faults, namely the F2 and F3 faults, are much less distinct in

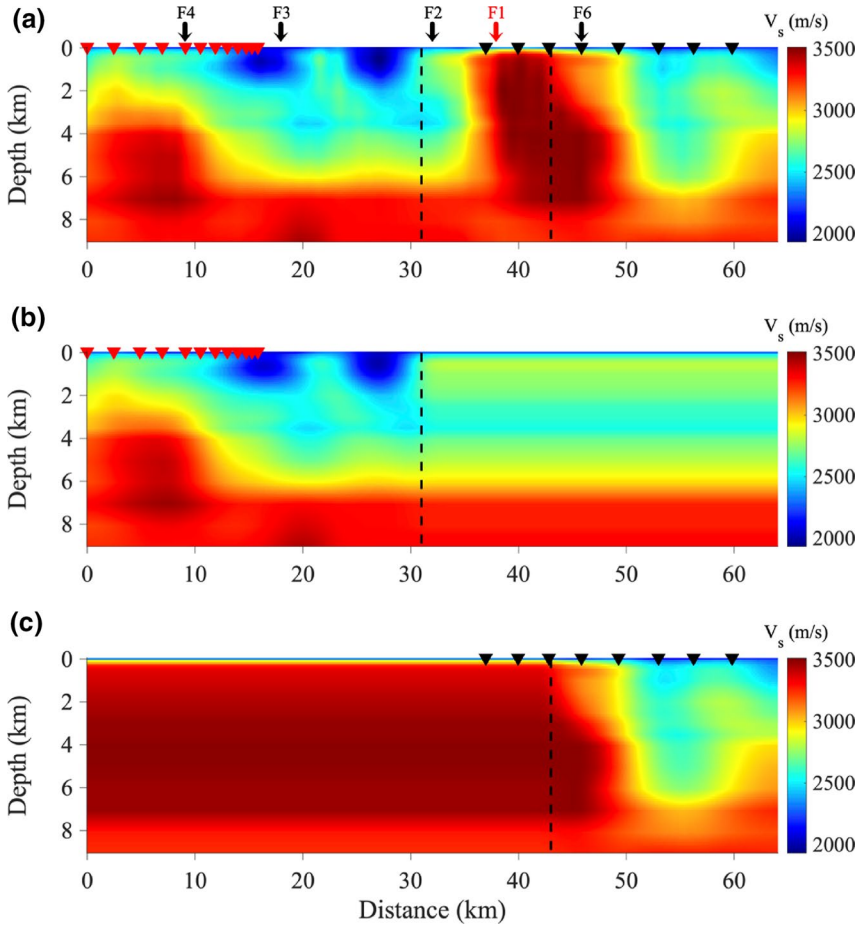


Figure 9. The same as Figure 8, except that there are 12 virtual sources (red inverted triangles) and 8 receivers (black inverted triangles) used in imaging the F1 fault.

the tomographic result, probably because those faults do not separate distinct lithological units. Therefore, we only image the former three faults with distinct velocity contrasts in this study. In the inverted velocity model in area B (Figure 11a), only one potential fault F15 is identified at distance around 10 km. To the west of the F15 fault lies the Cretaceous and Tertiary sediments with slow velocities, while the Cambrian and Ordovician ultrahigh pressure metamorphic rocks lying to the east of the F15 fault have much higher velocities. Some low-velocity intrusion is found within the metamorphic fault zone probably due to the strike-slip dislocation of the TFZ in the late Cretaceous era.

It is not trivial to interpret the accurate positions and the dipping angles of the subsurface faults from the smoothed tomographic results without clear fault imaging provided by TSW-RTM. The flood velocity models used for forward- and backward-propagating the source and receiver wavefields are obtained by extending the sediment and metamorphic blocks around the high velocity contrast in the subsurface model (dashed lines in Figures 8–11). We should emphasize the following facts when applying the TSW-RTM: (1) the velocity flood starts a short distance away from the potential fault plane, the exact position of which is not needed to know; (2) the flood velocity models used for forward- and backward-propagations do not include any velocity discontinuities around the fault zones.

For imaging the leftmost fault F4 in area A, 14 stations deployed between distance 11 and 18 km are selected as the virtual sources (red inverted triangles), and six stations deployed between distance 0 and 10 km are selected as the receivers (black inverted triangles) for the TSW-RTM imaging, as shown in Figure 8a. The flood velocity models for forward and backward propagations in the migration are illustrated in Figures 8b and 8c, respectively. Since it is difficult to estimate the source wavelet from the dispersive surface wave data,

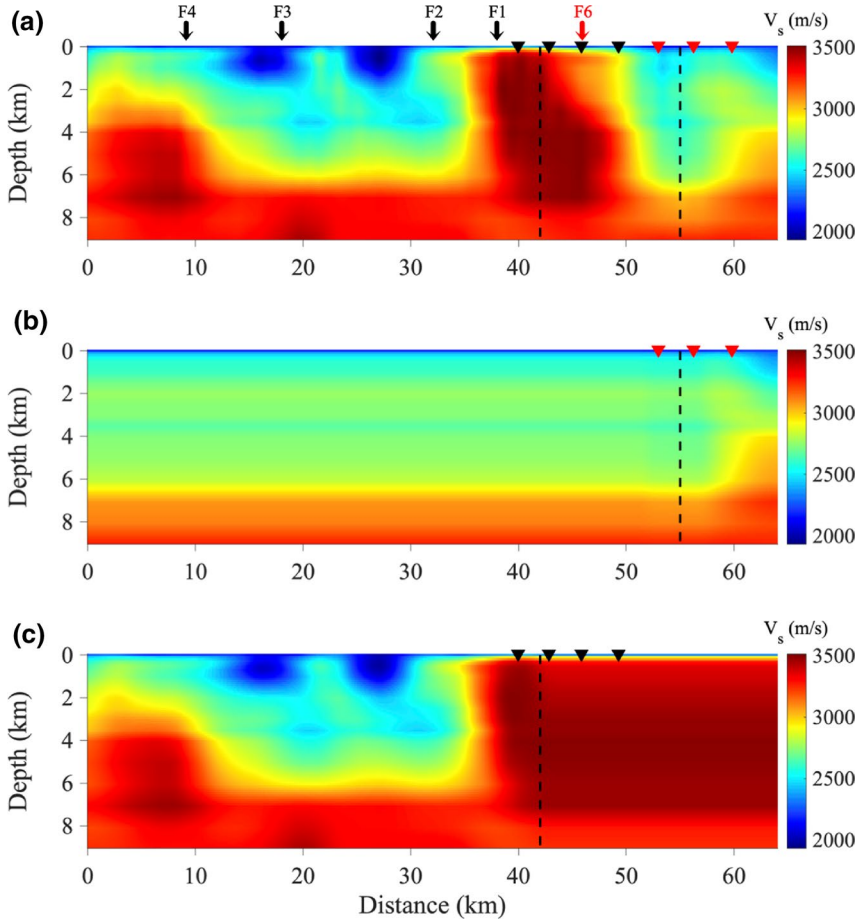


Figure 10. The same as Figure 8, except that there are three virtual sources (red inverted triangles) and four receivers (black inverted triangles) used in imaging the F6 fault.

we adopt the Ricker wavelet as the approximate source time function in the forward simulation with its central frequency (0.4 Hz) estimated from the spectrum of the virtual shot gathers. An image is obtained with Equation 1 by cross-correlating a forward wavefield, which is generated by injecting the estimated Ricker wavelet at a virtual shot position (red inverted triangle) with the back-propagated wavefield, which is generated by injecting the corresponding time-reversed virtual shot gather from all receivers together (black inverted triangles). After stacking images from all virtual shots, the final migrated image for the fault F4 is obtained, as shown in Figure 12a. The imaging result of the F4 fault shows great similarity with that from an active source seismic survey using traveltime tomography and poststack time migration (Qin et al., 2020), in terms of fault position and the dipping angle (~65°). We further adopt TSW-RTM to image another two faults F1 and F6 mapped on the surface (Figure 6b). Using the flood velocity models as given in Figures 9b and 9c, we obtain the image of the F1 fault, which is a high-angle fault (~75°) dipping to the north by northwest direction, as shown in Figure 12b. The image of the F6 fault using the flood velocity models as illustrated in Figures 10b and 10c is shown in Figure 12c, which is less steep (~60°) than the F1 fault, and dips approximately eastwards. To further validate our results, we compare the TSW-RTM fault images with the near-surface P-wave velocity model (Figure 13a) derived from the first-arrival traveltime tomography (Liu et al., 2015), and the S-wave velocity model (Figure 13b) derived from the ANT. In Figure 13, all the fault images of F1, F4, and F6 correlate nicely with the velocity discontinuities in P-wave velocity and the S-wave velocity models. The traditional first-arrival based traveltime tomography has limited depth penetration and does not yield reliable results at the depth more than 1 km (red dashed line in Figures 13b and 13c). In comparison, our TSW-RTM images shows clear fault delineation down to 4 km for all three faults. In general, the positions of the F1, F4, and F6 faults derived from TSW-RTM are well consistent with

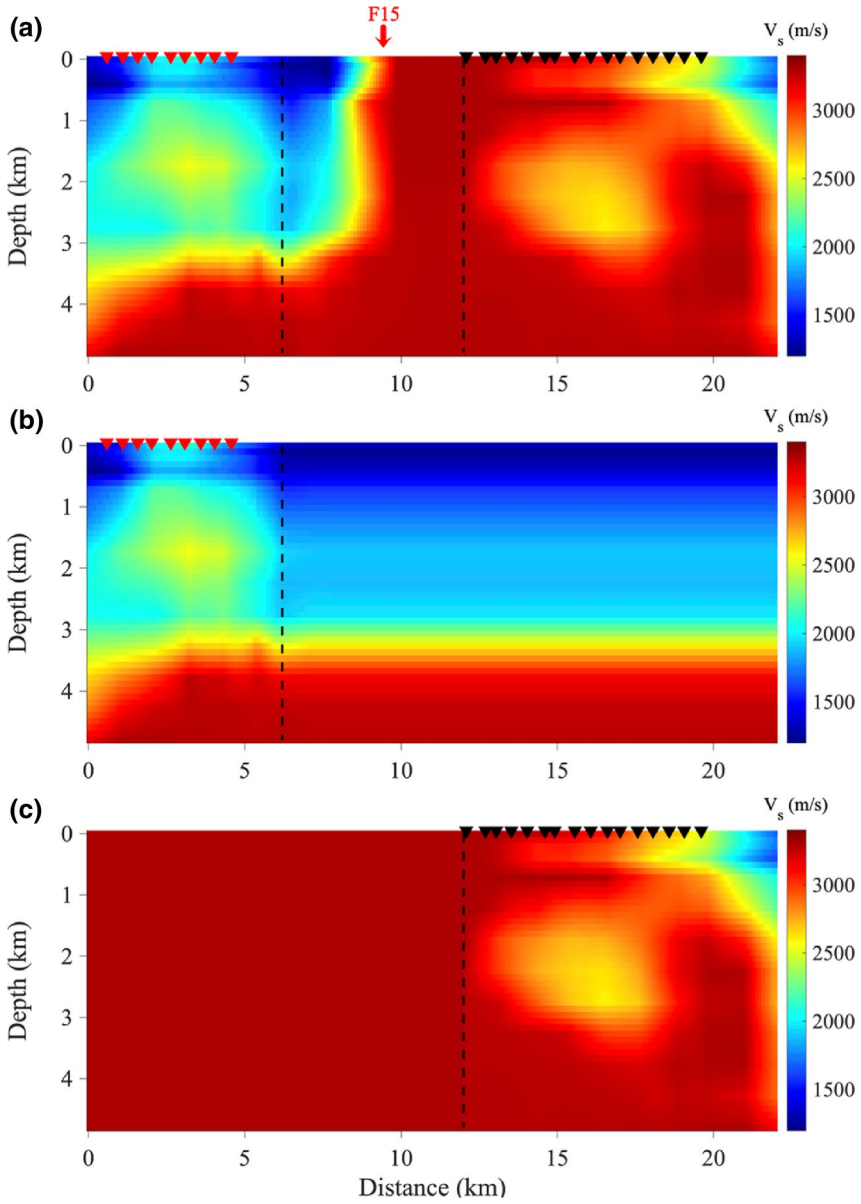


Figure 11. The same as Figure 8, except that there are 9 virtual sources (red inverted triangles) and 16 receivers (black inverted triangles) used in imaging the F15 fault in area B, with the position of F15 marked by the red arrow.

the surface map based on local outcrops and geological surveys, as well as the velocity models from active source tomography (Liu et al., 2015; Qin et al., 2020) and ANT. It is also noticed that the migrated image of fault F6 shows a deeper extension compared with faults F1 and F4. From the imaging perspective, the robustness of the imaging result decreases with depth due to weaker penetrated surface wave energy in our acquisition. Thus, the fault imaging result at greater depth may not always be as accurate as its shallower counterpart, though the horizontal energy normalization in the postprocessing has boosted the amplitude of the migrated image at greater depth. In our future research, the imaging uncertainty for deep structures will be further studied quantitatively.

In area B which is 250 km to the south of area A along the TFZ (Figure 6c), we also perform the proposed fault imaging method using the velocity flood models as shown in Figures 11b and 11c. Nine virtual sources and 16 receivers are used to image the F15 fault in area B. A Ricker wavelet with dominant frequency of 0.4 Hz is also adopted as the source wavelet in the forward simulation. After stacking contributions from

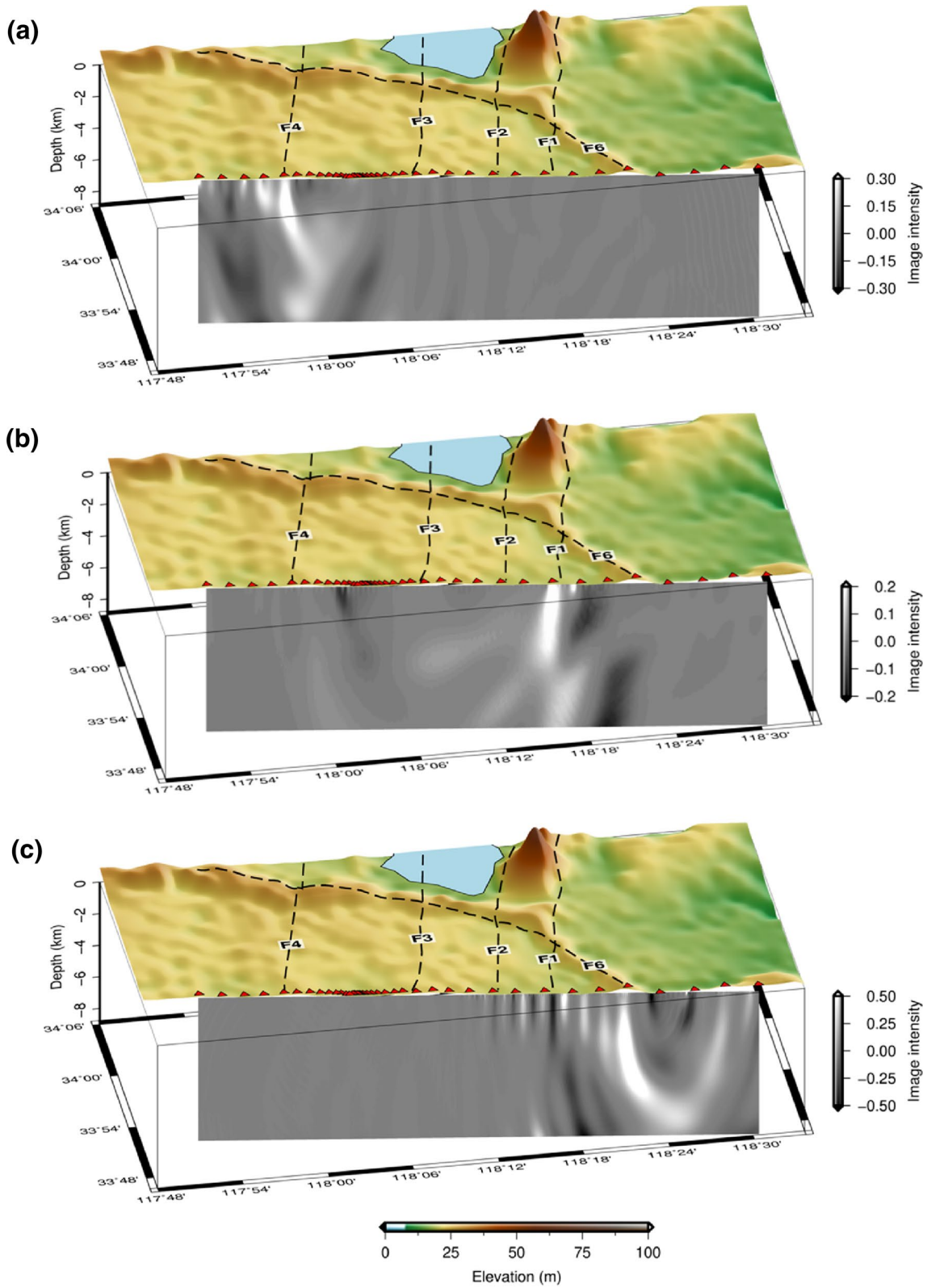


Figure 12. The TSW-RTM imaging results of fault planes of (a) F4, (b) F1, and (c) F6. The red triangles denote the deployed seismic nodes along the survey line. In the migrated images, the brightest areas with large amplitudes represent the fault planes, while other areas with relatively smaller amplitudes are migration artifacts.

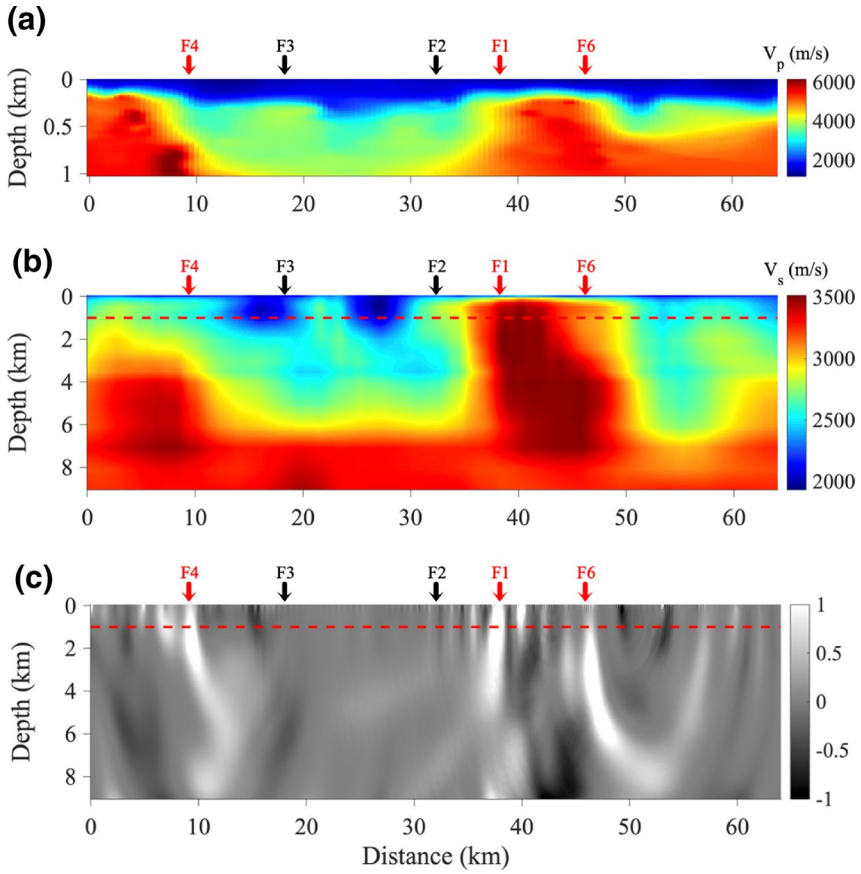


Figure 13. (a) The P-wave velocity model derived from the first-arrival traveltime tomography by an active-source seismic profiling in the same area. Note the vertical axis is exaggerated. (b) The S-wave velocity model derived from the ambient noise tomography along the linear array in area (a and c). The final imaging result of the Tanlu fault zone by combining the individual results of faults F1, F4, and F6 together in Figure 12. We balance the energy of different fault images for better illustration. The major structural faults imaged in this study with significant velocity contrast are indicated by red arrows whereas the minor faults not imaged are indicated by black arrows. The red dashed lines indicate the maximum penetration depth (1 km) of the active seismic profiling as shown in (a).

all source-receiver pairs, we obtain the fault image as shown in Figure 14. The position of the fault imaging result is highly consistent with its surface trace, and the F15 fault dips northwestwards with a high angle (~80°). Gu et al. (2019) study an adjacent segment of the TFZ to the north of area B, and their results also suggest that the F15 fault in this area dips northwestwards.

5. Discussion

Surface waves are dispersive in a heterogeneous medium with depth-dependent sensitivities, which render the imaging with surface waves different from that with body waves. To analyze the lateral imaging resolution, which characterizes the width of the migrated image for a fault plane, and the vertical imaging depth, which reflects the maximum trustworthy depth of the migrated image for a fault plane, the frequency of the wavefields and the velocity contrast across the fault discontinuity are two key controlling factors, which are analyzed in details in Appendix B for mathematical derivations and numerical test results. It is found that although the low-frequency part of a surface wave with better vertical penetration can be used for imaging the deeper segment of a fault, the lateral resolution of the image decreases owning to longer correlation lengths. On the other hand, the high-frequency part of a surface wave can image the shallower segment of a fault with relatively higher resolution, but it is incapable of imaging the deeper segment since its sensitivity decreases rapidly with depth. That is, the migrated images using the proposed TSW-RTM method gradually lose resolution with increasing depth.

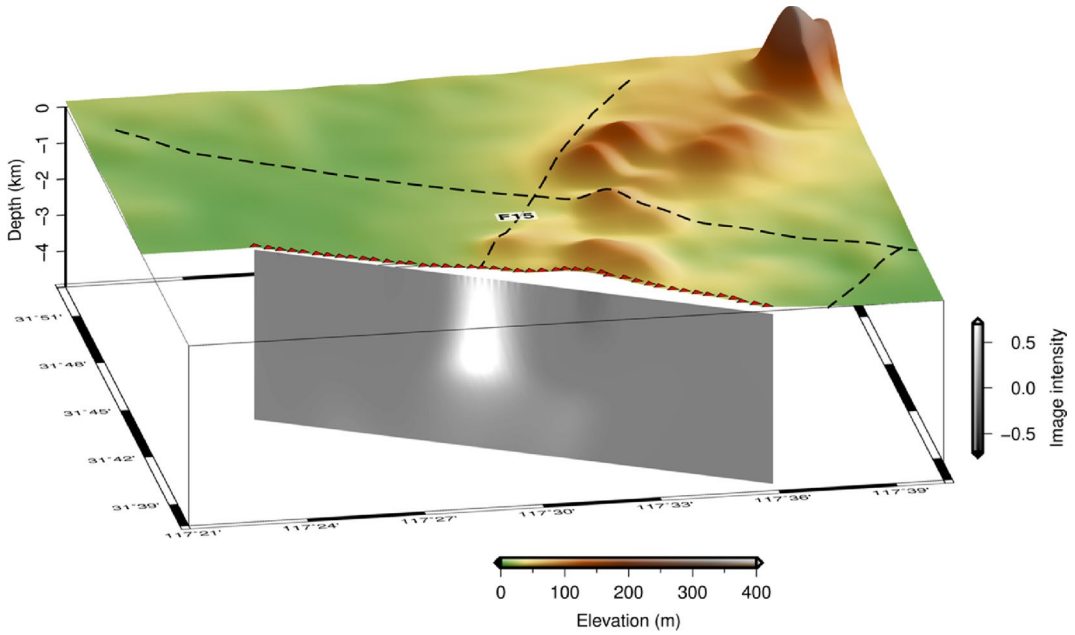


Figure 14. The Transmitted Surface Wave Reverse Time Migration imaging result of the F15 fault in area B. The red triangles denote the deployed seismic nodes along the survey line. The brightest areas with very large amplitudes represent the imaged F15 fault. The surface elevation along the linear array is minor, ranging from 5 to 65 m.

The proposed method also is flexible with the source-receiver configuration. Any receiver can be defined as the virtual source when a pair of receivers are used, as long as they are deployed at the two different sides of the target fault plane. It is recommended that the minimum distance of a receiver to the target fault plane should be designed accordingly by considering the desired imaging depth, resolution, and the frequency band of the used wavefield. A receiver deployed too close to a fault may not image the deep structures well with the transmitted surface waves for lack of penetration and illumination. Also, TSW-RTM does not have a strict requirement for the receiver intervals like the traditional reflection RTM, since each receiver is able to image the fault plane independently, though with a lower SNR and some artifacts.

Based on the two-way wave equation migration, the proposed TSW-RTM method can fully exploit the surface waves retrieved from ambient noise data sets. Our proposed method reveals more potentials of the ambient noise data by utilizing the transmission RTM principle to image the subsurface structures within the depth range that the surface waves are sensitive to. Both synthetic tests and field data studies demonstrate that the proposed method is capable of imaging fault planes with little a priori information, and it can delineate faults with higher resolution and certainty compared to the conventional tomography methods using surface wave dispersion curves. With the increased use of seismic nodes, our ambient noise fault imaging method is very useful in studying fault structures where the active-source data and earthquakes are not available. The proposed method can be further extended to image a fault surface in the three-dimensional space, which separates two geological blocks with distinct velocities. Though the proposed method is aimed to image fault planes, it is also applicable for imaging other interfaces in the subsurface, provided that the seismic velocity contrast across such an interface is not small and the dip angle is moderate to high.

Still, estimating the wavelet from the virtual shot gather retrieved from the ambient noise is nontrivial. In our current practice of the proposed TSW-RTM method, the source time function is assumed to be a Ricker wavelet with the dominant frequency estimated from the corresponding virtual shot gather, of which the time-reversed version is injected in the backpropagation. In our method, the lateral resolution of the fault image is naturally limited by the frequencies (0.2–1 Hz) of the reconstructed surface waves from the ambient noise cross-correlations. Still, the lateral resolution can be improved by deploying a denser array across the target fault. The migration deconvolution method (e.g., Schuster, 2017; Yu et al., 2006) can also be used to further improve the spatial resolution and suppress artifacts.

6. Conclusion

In this study, we have developed a novel ambient noise transmitted surface wave RTM method based on the velocity flood technique to image fault discontinuities. Using the transmitted surface waves, the proposed method in this study avoids retrieving body waves from the ambient noise cross-correlations, and directly utilizes the energetic surface waves for RTM. With the proposed method, the ambient noise data can be further exploited to provide a migrated image of the fault planes in addition to ANT model. This method is especially useful in areas that neither active nor passive seismic source data are available, though TSW-RTM is actually also applicable with actively or passively excited surface waves.

The synthetic tests demonstrate that our imaging method using transmitted surface waves can obtain clear images to delineate the faults only using a smooth migration velocity model in which positions and dips of the faults are ambiguous. The impact of errors in the migration velocity on the imaging quality is also discussed. Even in cases where the shapes of the faults are complicated, our method can still provide satisfactory imaging results. The proposed method is then applied to image two segments of the TFZ using the surface waves extracted from the ambient noise data sets recorded by linear arrays across the faults. Three major faults, namely F1, F4, and F6, in area A and the F15 fault in area B are all successfully imaged. Our fault imaging results agree well with previous studies in fault positioning and dip angles. We also analyze in detail how the wavefield frequency and the velocity contrast affect the lateral resolution and the vertical imaging depth. The proposed method extends the application of seismic ambient noise interferometry into the realm of RTM imaging. Therefore, we believe the proposed method will greatly benefit the applications of seismic ambient noise data, and potentially initiate and encourage new areas of research.

Appendix A

To better understand the virtual shot gathers retrieved from the ambient noise data, we simulate wavefields using the inverted velocity models shown in Figures 9a and 12a, respectively. The dominant frequency of the Ricker wavelet is selected based on the spectrum analysis of the virtual shot gather. Figure A1a shows the simulated shot gather excited at station eight using a Ricker wavelet with a dominant frequency of

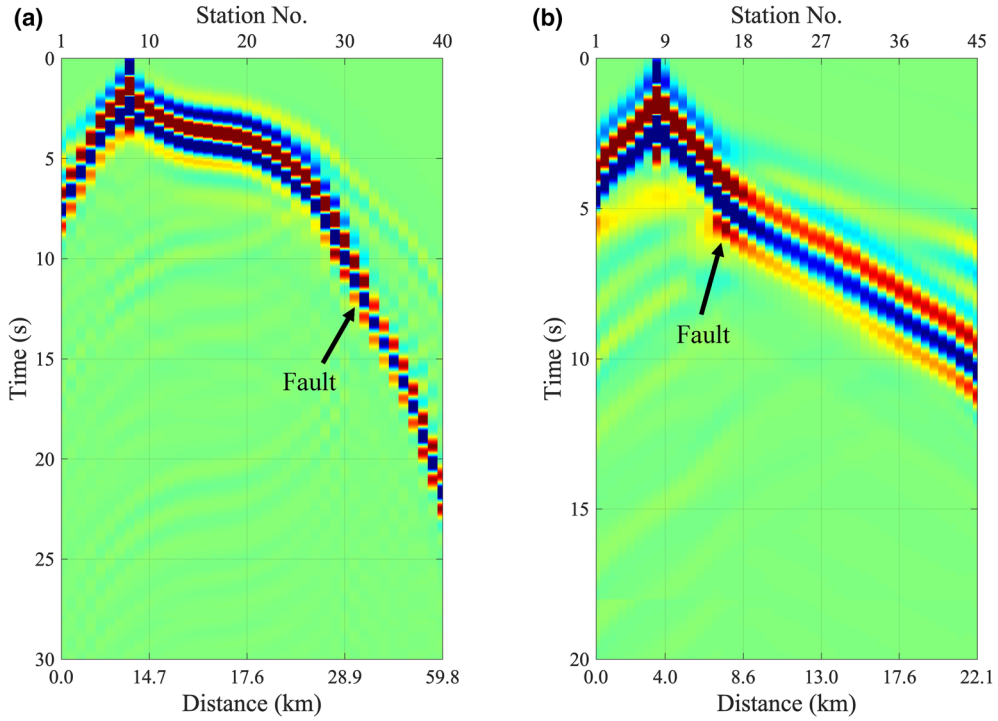


Figure A1. The simulated shot gathers in area A (a) and area B (b).

0.4 Hz. Compared with the preprocessed virtual shot gather as shown in Figure 7c, there are many similarities between the simulated and the observed virtual shot gathers, especially at the position of the suspected fault. We also compare the virtual shot gather from area B (Figure 8c) with the simulated one. The source is positioned at station 8 and a Ricker wavelet with a dominant frequency of 0.4 Hz is adopted as the source time function. The apparent velocity changes at around station 18 (Figure A1b), which is also observed in the virtual shot gather in Figure 8c.

Appendix B

To investigate the factors which influence the lateral resolution and the maximum imaging depth (the width and depth) in TSW-RTM, we consider an incident wavefield which propagates normal to an interface in the x–z plane, as shown in Figure B1. It should be noted that backpropagation of the time-reversed wavefield from the receivers is equivalent to reconstructing the transmitted wavefield forward in time. According to Equation 3, the incident surface wavefield P_i and the transmitted surface wavefield P_t can be expressed as:

$$P_i(x,z,\omega) = \xi_i(z,\omega)e^{ik_x x}, \tag{B1}$$

and

$$P_t(x,z,\omega) = \xi_t(z,\omega)e^{ik_x' x}, \tag{B2}$$

where $k_x = \omega/v_s$ and $k_x' = \omega / v_g$. In the proposed TSW-RTM method, the incident surface wavefield P_i and the transmitted surface wavefield P_t are correlated to obtain the migrated image; i.e.,

$$\boldsymbol{I}(x,z,\omega) \sim P_i P_t^* = \xi_i(z,\omega)\xi_t(z,\omega)e^{i\left(k_x - k_x'\right)x}. \tag{B3}$$

Therefore, the correlation length, which determines the horizontal resolution, is proportional to (Willacy and Kryvohuz, 2019):

$$l \propto \frac{1}{k_x - k_x'} = \frac{v_s v_g}{\omega\left|v_g - v_s\right|}. \tag{B4}$$

First, it is found the vertical imaging depth is related to the sensitivity kernels of the incident wavefield $\xi_i(z,\omega)$ and the transmitted wavefield $\xi_t(z,\omega)$, that is, the forward- and back-propagated surface wavefields in the TSW-RTM. For the lateral resolution, the correlation of the transmitted waves in the TSW-RTM imaging condition stretches the wiggles. The correlation length of the forward- and back-propagated wavefields, which is defined as a function of the overlap of the source and received wavefields (Chattopadhyay and McMechan, 2008) determine the horizontal resolving capability of the migrated image. As Equation B-4 indicates, the correlation length is related to the frequency ω of the wavefields and the velocity contrast v_g–v_s at the target fault plane in the proposed TSW-RTM method. That is to say, either surface wave with higher frequency or fault plane with higher velocity contrast has smaller correlation length and thus better lateral resolution.

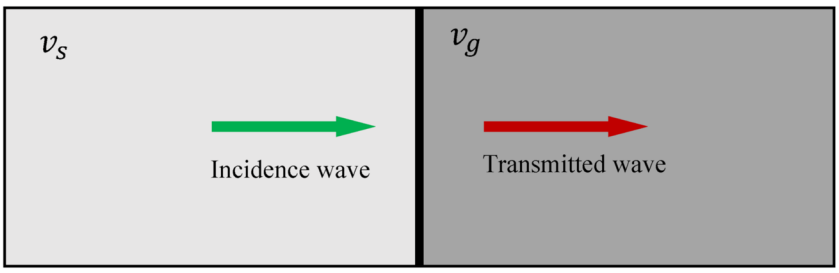


Figure B1. Schematic illustration of the incident and transmitted surface waves.

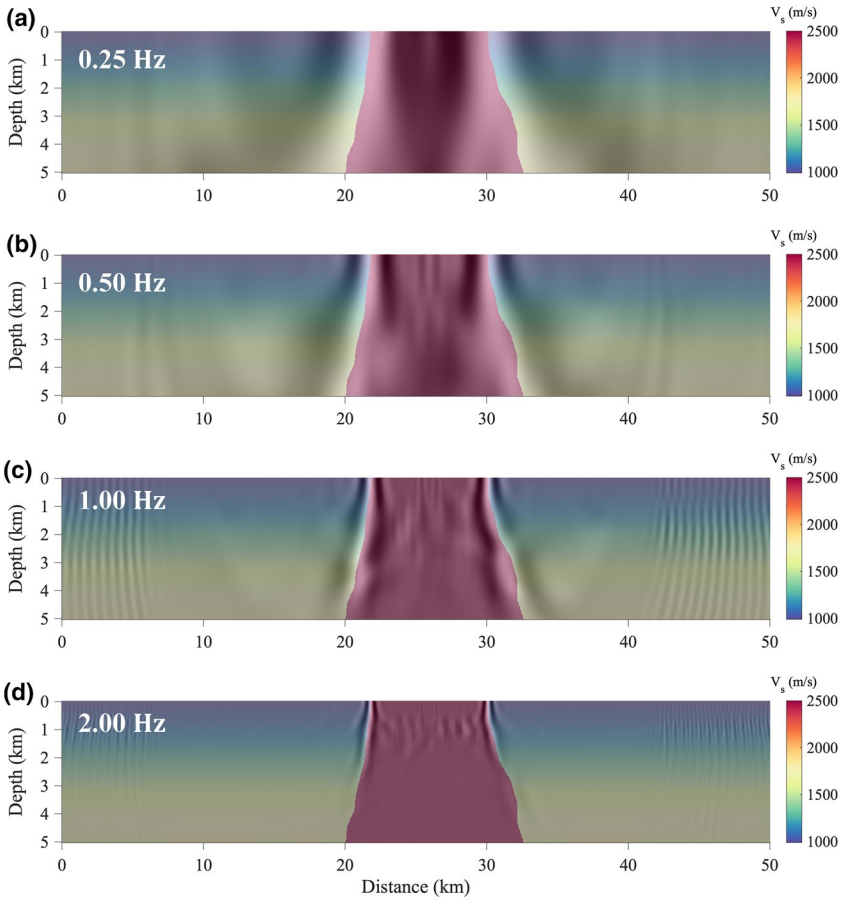


Figure B2. Imaging results using sources with four different dominant frequencies: (a) 0.25 Hz, (b) 0.50 Hz, (c) 1.00 Hz, and (d) 2.00 Hz.

We conduct a few more synthetic tests to further elaborate the dependence of correlation length on the wave-field frequency and the velocity contrast. The first group tests compare the imaging results using sources with different dominant frequencies. The velocity of the metamorphic rock is 2,500 m/s in the tests, and the target imaging depth is about 5 km for the faults. The dominant frequencies of the injected source wavelets are 0.25, 0.50, 1.00, and 2.00 Hz with the frequency bandwidth ranging from 0.125 to 0.375 Hz, 0.25 to 0.75 Hz, 0.50 to 1.50 Hz, and 1.00 to 3.00 Hz, respectively. The same source wavelet in each test is also used to generate the corresponding virtual shot gathers which are later time-reversed and injected at the receivers. The imaging result using varying sources are given in Figure B2. In Figures B2a and B2b, it is found that the lateral resolutions of the fault images are relatively low when using the 0.25 and 0.50 Hz source wavelets, and the interpretation of the fault positioning is thus difficult. In comparison, the imaging result using a 1.0 Hz source wavelet with bandwidth of 0.50–1.50 Hz (Figure B2c) achieves a good balance between the lateral resolution and the vertical penetration. When using a wavelet with even higher dominant frequency (2.0 Hz), the shallow segments of the faults are clearly resolved whereas the deeper segments lack illumination from the surface waves and thus are poorly imaged, as shown in Figure B2d.

We further test the dependence of the lateral resolution of the image on the velocity contrast across the fault surface. The velocities of the metamorphic rocks are assumed to be 2,000, 2,500, and 3,000 m/s to generate different contrasts against the sediments in three tests. A wavelet with dominant frequency of 1.0 Hz is used in all cases. The lateral resolution improves noticeably from Figures B3a to B3b with the velocity contrast at the fault plane increasing. While the improvement from Figures B3b to B3c is not obvious, as the lateral resolution has a diminishing dependence on the increased velocity contrast. That is, in the high contrast cases, the dominant frequency of the surface wave has a more pronounced influence on the lateral resolution.

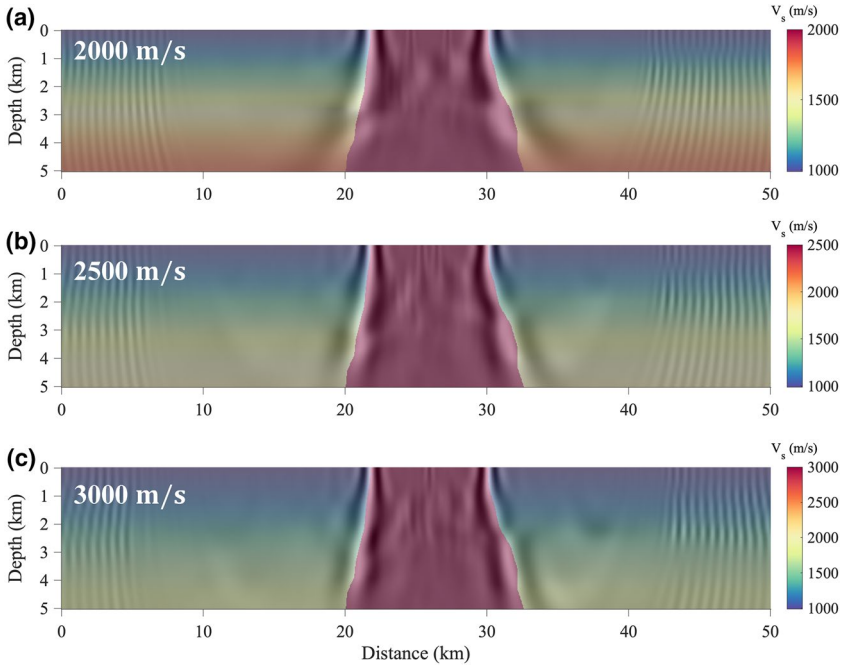


Figure B3. Imaging results of three fault zone models with different velocity contrasts across the faults. The Ricker wavelet with a dominant frequency of 1 Hz is adopted in the tests.

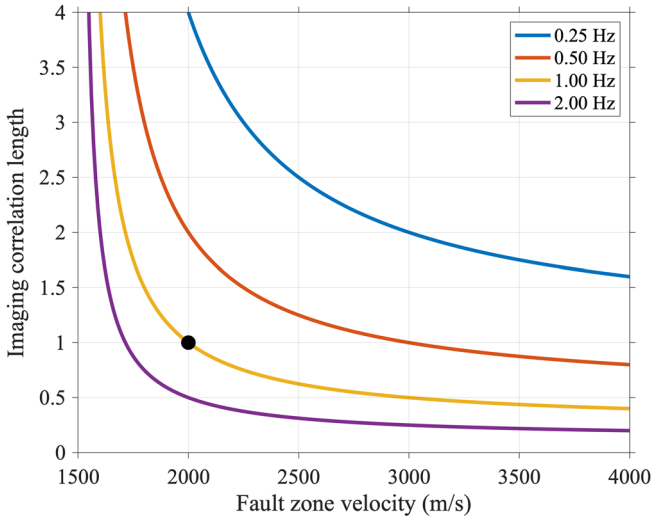


Figure B4. The relationship between the fault zone velocity and the imaging correlation length at four different frequencies (0.25, 0.50, 1.00, and 2.00 Hz). The sediment velocity is assumed to be a constant of 1,500 m/s for the sake of discussion. The imaging correlation lengths are normalized to the case where the fault zone velocity is 2,000 m/s and the frequency of the wavefield is 1.0 Hz (black-solid circle).

Apart from the imaging examples presented above, the quantitative analysis of the dependence of the lateral imaging resolution on the velocity contrast is also performed. In Figure B4, we show the relationship between the fault zone velocity and the imaging correlation length which determines the lateral imaging resolution based on Equation B4. The sediment velocity (v_s) is assumed to be a constant of 1,500 m/s for the sake of discussion. Then, the imaging correlation lengths are calculated for varying fault zone velocities (v_g) ranging from 1,500 to 4,000 m/s at four different frequencies (0.25, 0.50, 1.00, and 2.00 Hz). For com-

parison, the imaging correlation length for the case where $v_g = 2{,}000$ m/s and $f = 1.0$ Hz (black-solid circle) is set to unit, while other correlation lengths are normalized to this case. In Figure 4b, it is noticed that the imaging correlation length decreases with increasing fault zone velocity, and gradually levels off. Also, the higher the used frequency is, the short the correlation length becomes. Equation B4 and Figure B4 provide the guidance to estimate the lateral imaging resolution for interfaces with different velocity contrasts using surface waves of different frequencies.

Data Availability Statement

The calculated CCFs, the velocity profiles, and the fault imaging results used in this study can be accessed via https://zenodo.org/record/4040387#.X2hS-5MzaMJ.

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